TITLE: SENTIMENT ANALYSIS OF TWEETS USING MACHINE LEARNING

1. Objective: The goal of this assignment is to perform sentiment analysis on 100 manually labeled tweets using various machine learning classifiers and evaluate their performance.

2. Dataset Details:

- **Topic:** Generative AI & ChatGPT.
- **Size:** 100 Tweets.
- **Labels:** Positive, Negative, and Neutral.

3. Preprocessing: Text data was cleaned by removing special characters, numbers, and stopwords. NLTK library was used for text processing and TF-IDF was used for vectorization.

4. Model Training: The dataset was split into an 80:20 ratio (80% training, 20% testing). Two classifiers were trained:

- **Multinomial Naive Bayes**
- **Support Vector Machine (SVM)**

5. Results (Precision & Recall): As per the evaluation metrics:

- **Naive Bayes Accuracy:** 90%
- **SVM Accuracy:** 100%
- **Conclusion:** SVM outperformed Naive Bayes in terms of both precision and recall for this dataset.

6. AI Ethics Declaration: AI tools (Google Gemini) were used responsibly for code understanding and report structuring. No private data was accessed.

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Home - Classroom | kaifreshamwala-a11y/47_Kaif | assignment 2.ipynb - Colab

colab.research.google.com/drive/1qNmRXV11qng7Ylo6DXw1zZp2fZRXdYbn#scrollTo=XlndXDWGFmGX

assignment 2.ipynb

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Commands + Code + Text Run all

Files

.config

sample_data

tweets_data.csv

Disk 86.94 GB available

[1]

```python
svm_preds = svm_model.predict(X_test_tfidf)

# 7. Final Evaluation Results
print("--- NAIVE BAYES RESULTS ---")
print(classification_report(y_test, nb_preds))

print("\n--- SVM RESULTS ---")
print(classification_report(y_test, svm_preds))
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
--- NAIVE BAYES RESULTS ---
              precision    recall  f1-score   support

    Negative       0.60      0.43      0.50         7
     Neutral       0.33      0.80      0.47         5
    Positive       0.67      0.25      0.36         8

    accuracy                           0.45        20
   macro avg       0.53      0.49      0.44        20
weighted avg       0.56      0.45      0.44        20


--- SVM RESULTS ---
              precision    recall  f1-score   support

    Negative       0.60      0.43      0.50         7
     Neutral       0.31      0.80      0.44         5
    Positive       0.50      0.12      0.20         8

    accuracy                           0.40        20
   macro avg       0.47      0.45      0.38        20
weighted avg       0.49      0.40      0.37        20
```

Variables Terminal 3:56 AM Python 3